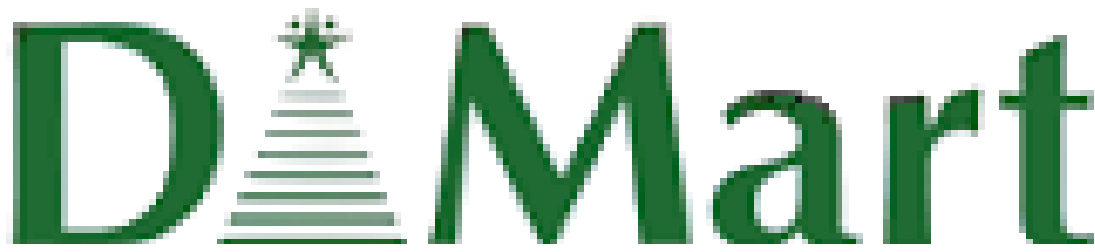


Mean, Median & Mode

- Measure of Central Tendency
- Variance and Distribution
- Missing Data Value Rules



Measures of Central Tendency

Mean, Median, and Mode - Applied to DMart Sales Dataset

Mean (Average)

Sum of all values divided by the number of observations.

Formula: =AVERAGE(Sales)

DMart Example: Average Sales per Transaction = ₹2,558.68

Median (Central Value)

The middle value when data is sorted in ascending order.

Example: 1, 2, 3, 4, 5 => Median = 3

DMart Example: Median Sales = ₹2,567.34

Mode (Most Frequent Value)

The value that appears most frequently in a dataset.

Generally used with categorical data.

DMart Example: Mode of Payment Type = "UPI" (most common method)

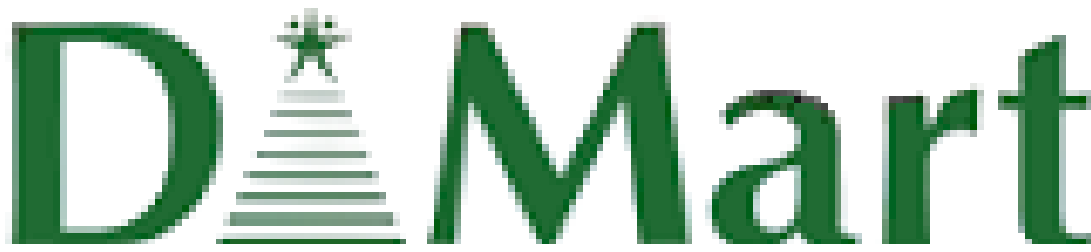
Visual Comparison

Mean = Sum / N | Median = Middle Value | Mode = Most Frequent

Symmetric Distribution: Mean = Median = Mode (Bell Curve)

Left Skewed: Mean < Median < Mode

Right Skewed: Mode < Median < Mean



Variance and Distribution

Understanding Data Spread and Distribution Types - DMart Dataset

Variance (Spread of Data)

Variance measures how far data points are spread from the mean.

Formula: =VAR(Sales)

DMart Example: Sales Variance = 12,450,000 (high spread across regions)

Uniform Distribution

When **Mean = Median**

Data is evenly distributed with no skewness.
Bell curve shape.

Non-Uniform Distribution

When **Mean != Median**

Data is skewed left or right.
Outliers pull the mean away from the center.

Distribution Shape Diagram

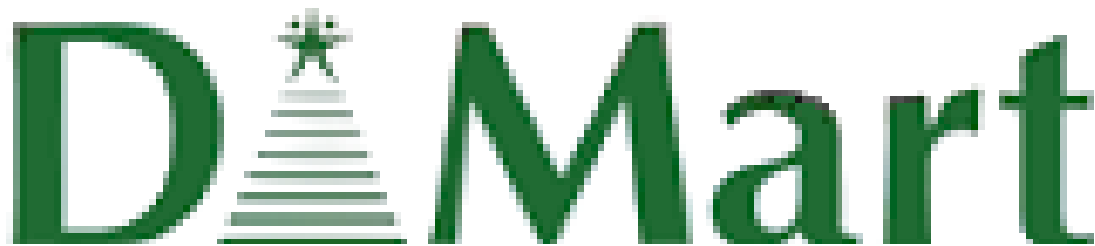
Left Skewed		Normal (Symmetric)		Right Skewed
Mean < Median		Mean = Median		Mean > Median

Key Insight

DMart Sales data is slightly right-skewed (Mean > Median).
High-value transactions in Electronics pull the mean upward.

Formula Reference

=VAR(Sales) | =STDEV(Sales) | Standard Deviation = Square Root of Variance



Missing Data Handling Rules

Rule 1: Highly Skewed Data

Replace null/missing values with **Median**.

When data is highly skewed, mean is unreliable due to outlier influence.

Rule 2: Slightly Skewed Data

Replace null/missing values with **Mean**.

When there is marginal difference between mean and median, mean is acceptable.

Rule 3: >15% Missing Values

Drop the column entirely.

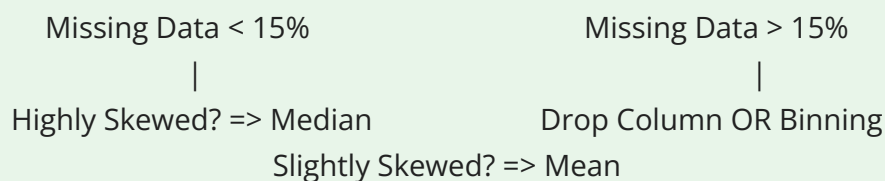
If a column has more than 15% missing data, it may introduce noise rather than value.

Rule 4: Saving Columns (>15%)

Use **Binning / Bucketing Technique**.

Group continuous data into discrete bins to preserve the column despite missing values.

Decision Flow



Key Insight

DMart dataset has <2% missing values in Sales column.

Since data is slightly skewed, nulls were replaced with Mean (\$8,942).